

Research Proposal Template

1. Proposal Metadata

Proposal title:

[Short, specific title]

Research direction:

[direction_id / topic]

Prepared by:

[Ariadne / human / team]

Date:

[YYYY-MM-DD]

Status:

Draft / Under review / Approved / Rejected / Paused

Linked literature review:

[literature_review_id]

Linked Library topic:

[topic_id]

2. Executive Summary

This proposal investigates:

[One-sentence statement of the research problem.]

The proposed contribution is:

[One-sentence statement of the intended contribution.]

The central hypothesis is:

[One-sentence hypothesis.]

The proposal is motivated by a gap in the literature:

[Briefly state the gap.]

3. Background and Motivation

Research context

[Summarize the area and why it matters.]

Problem

[Describe the specific problem this research addresses.]

Why current approaches are insufficient

[Explain limitations of existing methods, systems, datasets, or evaluations.]

Why this matters now

[Explain timeliness: new models, new agent systems, new benchmarks, emerging failures, research trend, etc.]

4. Literature Review Summary

The literature review identified the following relevant clusters:

1. [Cluster 1]
2. [Cluster 2]
3. [Cluster 3]

Key prior work includes:

Paper	Contribution	Relevance
[Paper 1]	[Contribution]	[Why relevant]
[Paper 2]	[Contribution]	[Why relevant]
[Paper 3]	[Contribution]	[Why relevant]

The main gap is:

[State the research gap clearly.]

5. Research Gap

Gap statement

[Clearly state the gap in one paragraph.]

Evidence for the gap

The gap is supported by:

- [Evidence 1 from literature]
- [Evidence 2 from literature]
- [Evidence 3 from literature]

Why the gap is researchable

This gap can be investigated because:

- [Reason 1]
- [Reason 2]
- [Reason 3]

Why the gap is not already solved

[Explain why existing work does not fully address it.]

6. Research Question

Primary research question:

[Main research question]

Secondary research questions:

1. [Sub-question 1]
 2. [Sub-question 2]
 3. [Sub-question 3]
-

7. Hypothesis

Primary hypothesis:

[Hypothesis]

Expected outcome:

[Expected measurable effect or result.]

Alternative hypothesis:

[Alternative explanation or competing hypothesis.]

Null or failure condition:

[What result would indicate the hypothesis is unsupported?]

8. Proposed Contribution

This research aims to contribute:

1. [Contribution 1]
2. [Contribution 2]
3. [Contribution 3]

Example contribution types:

- new method
 - new benchmark
 - new evaluation framework
 - new dataset
 - new system architecture
 - new empirical finding
 - new analysis of failure modes
 - new evidence-grounding mechanism
-

9. Methodology

Proposed approach

[Describe the method or system to be built/tested.]

Baselines

The proposed approach will be compared against:

Baseline	Reason for inclusion
[Baseline 1]	[Reason]
[Baseline 2]	[Reason]
[Baseline 3]	[Reason]

Datasets

Dataset	Source	Purpose	Mimir status
[Dataset 1]	[Source]	[Use]	Approved / Pending / Blocked
[Dataset 2]	[Source]	[Use]	Approved / Pending / Blocked

Metrics

Metric	What it measures	Why it matters
[Metric 1]	[Description]	[Reason]
[Metric 2]	[Description]	[Reason]
[Metric 3]	[Description]	[Reason]

10. Experimental Plan

Experiment 1: [Experiment title]

Purpose:

[What this experiment tests.]

Input:

[Data, models, prompts, systems, etc.]

Method:

[Experimental procedure.]

Expected result:

[Expected measurable outcome.]

Success criteria:

[What would count as success.]

Failure / kill condition:

[What would cause this experiment to be stopped or deprioritized.]

Experiment 2: [Experiment title]**Purpose:**

[What this experiment tests.]

Input:

[Data, models, prompts, systems, etc.]

Method:

[Experimental procedure.]

Expected result:

[Expected measurable outcome.]

Success criteria:

[What would count as success.]

Failure / kill condition:

[What would cause this experiment to be stopped or deprioritized.]

11. Claim Goals

The proposal aims to test the following claim goals:

Claim goal	Expected evidence	Success criteria	Kill condition
[cg_001]	[Evidence needed]	[Success]	[Kill condition]
[cg_002]	[Evidence needed]	[Success]	[Kill condition]

Example claim goal:

```
{
  "claim_goal_id": "cg_007",
  "target_claim": "Context graphs reduce unsupported claims in AI-generated research summaries.",
  "expectation": "At least 20% reduction compared with document-only RAG.",
  "evidence_needed": [
    "baseline comparison",
    "multi-dataset evaluation",
    "critic review",
    "human or calibrated judge validation"
  ],
}
```

```
"kill_conditions": [  
  "improvement below 10%",  
  "cost exceeds 3x baseline",  
  "effect disappears under human evaluation"  
]  
}
```

12. Expected Results

Expected findings:

1. [Expected finding 1]
2. [Expected finding 2]
3. [Expected finding 3]

Expected artifacts:

- [Artifact 1]
- [Artifact 2]
- [Artifact 3]

13. Risks and Limitations

Risk	Impact	Mitigation
[Risk 1]	Low / Medium / High	[Mitigation]
[Risk 2]	Low / Medium / High	[Mitigation]
[Risk 3]	Low / Medium / High	[Mitigation]

Common risks:

- weak novelty
 - weak baseline
 - insufficient evaluation
 - LLM-judge bias
 - dataset leakage
 - unclear dataset license
 - insufficient replication
 - system engineering without research contribution
 - too expensive to run
 - results not statistically meaningful
-

14. Resources Required

Data

[Datasets, papers, repositories, logs, generated data.]

Compute

[GPU, CPU, memory, storage, API budget.]

Agents involved

- Ariadne
- Mimir
- Planner
- Researchers
- Experiments
- Critic
- Gate
- Quartermaster

Tools

[Libraries, models, databases, evaluation frameworks.]

15. Timeline

Phase	Description	Output	Estimated duration
Literature review	[Description]	Literature review	[Duration]
Proposal approval	[Description]	Approved proposal	[Duration]
Experiment design	[Description]	Experiment plan	[Duration]
Implementation	[Description]	Code and runs	[Duration]
Evaluation	[Description]	Results	[Duration]
Review	[Description]	Critic and Gate verdicts	[Duration]
Writing	[Description]	Paper draft	[Duration]

16. Approval Gates

Before experiments begin:

- ✓ Literature review completed
- ✓ Research gap approved by Ariadne
- ✓ Datasets approved by Mimir
- ✓ Experiment plan approved by Planner
- ✓ Compute budget approved by Quartermaster

Before claim promotion:

- ✓ Results produced
- ✓ Evidence linked
- ✓ Critic review completed
- ✓ Mimir trust checks passed
- ✓ Gate promotion criteria passed

Before publication:

- ✓ Claims are paper-ready
- ✓ Evidence chains complete
- ✓ Limitations documented
- ✓ Reproducibility package prepared
- ✓ Human approval granted

17. Expected Paper Shape

Potential paper title:

[Possible title]

Potential paper contribution statement:

[Contribution statement]

Potential paper sections:

Abstract
Introduction
Related Work
Method
Experiments
Results

Limitations
Conclusion

18. Proposal Verdict

Recommendation:

Approve / Revise / Reject / Pause

Reason:

[Explain the verdict.]

Next step:

[Literature review / experiment plan / request to Mimir / additional validation / human review]